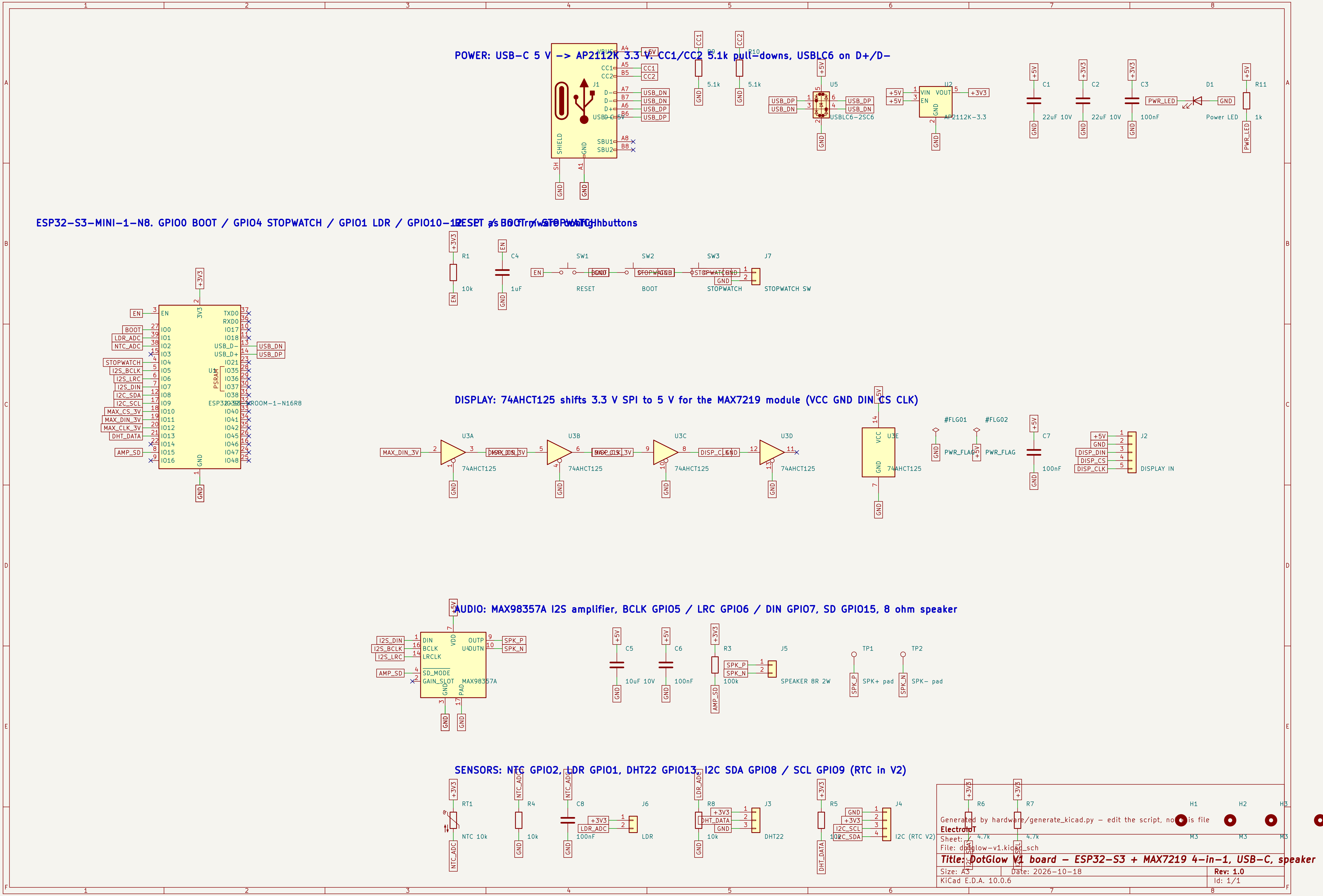
J1
VBUS
A4
+5V
CC1
A5
CC1
CC2
B5
CC2
D-
A7
USB_DN
D-
B7
USB_DN
D+
A6
USB_DP
USB_DP
B6
USB_DP
USB-C 5V
SBU1
A8
SBU2
B8
SHIELD
GND
SH
A1
GND
GND
CC1
R9
5.1k
GND
CC2
R10
5.1k
GND
+5V
U5
USB_DP
USB_DN
USB_DP
USB_DN
USBLC6-2SC6
GND
U2
+5V
+5V
VIN
VOUT
EN
GND
+3V3
AP2112K-3.3
GND
+5V
C1
22uF 10V
GND
+3V3
C2
22uF 10V
GND
+3V3
C3
100nF
GND
D1
PWR_LED
GND
Power LED
+5V
R11
1k
PWR_LED
+3V3
U1
EN
3V3
IO0
IO1
IO2
IO3
IO4
IO5
IO6
IO7
IO8
IO9
IO10
IO11
IO12
IO13
IO14
IO15
IO16
TXD0
RXD0
IO17
IO18
USB_D-
USB_D+
IO21
IO35
IO36
IO37
IO38
IO39
IO40
IO41
IO42
IO45
IO46
IO47
IO48
PSRAM
ESP32-S3-WROOM-1-N16R8
GND
EN
BOOT
LDR_ADC
NTC_ADC
STOPWATCH
I2S_BCLK
I2S_LRC
I2S_DIN
I2C_SDA
I2C_SCL
MAX_CS_3V
MAX_DIN_3V
MAX_CLK_3V
DHT_DATA
AMP_SD
USB_DN
USB_DP
GND
+3V3
R1
10k
EN
EN
C4
1uF
GND
SW1
EN
BOOT
RESET
SW2
STOPWATCH
BOOT
SW3
STOPWATCH
GND
STOPWATCH
J7
GND
STOPWATCH SW
U3A
MAX_DIN_3V
DISP_DIN
74AHCT125
GND
U3B
MAX_CS_3V
DISP_CS
74AHCT125
GND
U3C
MAX_CLK_3V
DISP_CLK
74AHCT125
GND
U3D
GND
74AHCT125
GND
+5V
U3E
VCC
GND
74AHCT125
GND
#FLG01
GND
PWR_FLAG
#FLG02
+5V
PWR_FLAG
+5V
C7
100nF
GND
J2
+5V
GND
DISP_DIN
DISP_CS
DISP_CLK
DISPLAY IN
+5V
U4
I2S_DIN
I2S_BCLK
I2S_LRC
AMP_SD
DIN
BCLK
LRCLK
SD_MODE
GAIN_SLOT
VDD
OUTP
OUTN
GND
PAD
SPK_P
SPK_N
MAX98357A
GND
GND
+5V
C5
10uF 10V
GND
+5V
C6
100nF
GND
+3V3
R3
100k
AMP_SD
J5
SPK_P
SPK_N
SPEAKER 8R 2W
TP1
SPK_P
SPK+ pad
TP2
SPK_N
SPK- pad
+3V3
RT1
NTC 10k
NTC_ADC
NTC_ADC
R4
10k
GND
NTC_ADC
C8
100nF
GND
J6
+3V3
LDR_ADC
LDR
LDR_ADC
R8
10k
GND
J3
+3V3
DHT_DATA
GND
DHT22
+3V3
R5
10k
DHT_DATA
J4
GND
+3V3
I2C_SCL
I2C_SDA
I2C (RTC V2)
+3V3
R6
4.7k
I2C_SDA
+3V3
R7
4.7k
I2C_SCL
H1
M3
H2
M3
H3
M3

POWER: USB-C 5 V -> AP2112K 3.3 V. CC1/CC2 5.1k pull-downs, USBLC6 on D+/D-

ESP32-S3-MINI-1-N8. GPIO0 BOOT / GPIO4 STOPWATCH / GPIO1 LDR / GPIO10-12 as firmware config

RESET / BOOT / STOPWATCH buttons

DISPLAY: 74AHCT125 shifts 3.3 V SPI to 5 V for the MAX7219 module (VCC GND DIN CS CLK)

AUDIO: MAX98357A I2S amplifier, BCLK GPIO5 / LRC GPIO6 / DIN GPIO7, SD GPIO15, 8 ohm speaker

SENSORS: NTC GPIO2, LDR GPIO1, DHT22 GPIO13, I2C SDA GPIO8 / SCL GPIO9 (RTC in V2)

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**ElectroIoT**

Sheet: /

File: dotglow-v1.kicad_sch

***Title: DotGlow V1 board – ESP32-S3 + MAX7219 4-in-1, USB-C, speaker***

Size: A3 Date: 2026-10-18 **Rev: 1.0**

KiCad E.D.A. 10.0.6 Id: 1/1